

# AI-Powered Dental Caries Detection

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**Abstract-** Tooth decay, or dental caries, is a prevalent global health issue that underscores the need for early detection and diagnosis to effectively prevent serious dental problems. Our research, "AI Powered Dental Caries Detection" seeks to address this challenge by utilizing Artificial Intelligence (AI) for early detection of dental caries. This research focuses on analyzing intraoral images taken via smartphone cameras. Our research harnesses state-of-the-art deep learning (DL) object detection models such as YOLO (You Only Look Once), EfficientDet, MobileNet and resNet-50 to deliver a highly effective solution for detecting dental caries in the early stages. In addition to improving dental care's effectiveness, this kind of technology also lowers costs and increases accessibility, especially for people living in low- and middle-income areas where traditional dental services could be out of reach. Our research goal is to make proactive management of oral health possible, which will eventually enhance general health and quality of life. Our research indicates that ResNet-50 is the most effective model, while MobileNet serves as a strong alternative. The study introduces YOLOv8 and ResNet-50 for dental caries detection in mobile app. These findings have substantial implications for enhancing dental care practices and potentially increasing accessibility to early detection across various demographics.

**keywords—**Dental Caries, Deep Learning Models, Object Detection

## I. INTRODUCTION

Dental caries, commonly known as tooth decay, is a prevalent oral health issue that affects people of all ages worldwide as shown in Fig. 1. It is a major concern for public health due to its widespread nature and the significant impact it has on individuals' overall well-being. We are developing a groundbreaking mobile application titled "Dental Caries

Detection". The early ways of detection are X-rays, visual examinations and these ways are not just costly but also, they are time consuming and painful. Firstly, patient has to go to Dentist and book an appointment and then wait for hours to get checked. This problem is particularly acute in populous and developing countries like Pakistan, where obtaining a dental appointment can be especially challenging due to the high demand for services. And what if there is no caries it was just an ordinary pain. The patient's money and time got wasted. But in this Research paper we are introducing a system for the early detection. This revolutionary app is designed to empower users by providing them with the ability to identify initial dental caries effectively. The application leverages advanced computer vision algorithms, such as YOLO (You Only Look Once) [1], EfficientDet [2], MobileNet [3] and resNet-50 [4] to assess images of user's teeth. These models are trained on a public caries dataset to ensure accurate and reliable assessments. This app analyzes intraoral images in natural light, ensuring accessibility without special equipment. By simply taking a picture with their smart phone camera and uploading it to the app, users can receive real-time assessments of their dental health status. The application not only provides an initial assessment but also offers a comprehensive suite of features aimed at promoting optimal oral health. Users can browse profiles of nearby dental professionals, view their availability in real-time, and book appointments directly within the app. This streamlines the process of accessing dental care, eliminating the need for lengthy waiting periods and ensuring prompt attention to dental issues. The user-friendly interface of our application ensures that it is accessible to individuals of all ages and technical backgrounds. Whether at home or on-the-go, users can easily navigate through the app, capture dental images, and access relevant information and services with minimal effort. By integrating cutting-edge technology with user-centric design principles, the "Dental Caries Detection" application aims to revolutionize dental care delivery. This innovative solution empowers users to take proactive steps

towards maintaining optimal oral health, ultimately improving their overall well-being and quality of life.

## II. RELATED WORK

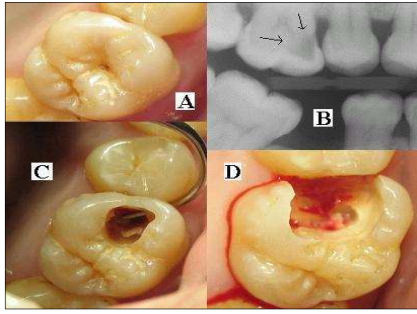


Figure 1: **Dental caries:** (A) A small spot of decay visible on the surface of a tooth. (B) The radiograph reveals an extensive region of demineralization within the dentin (arrows). (C) A hole is discovered on the side of the tooth at the beginning of decay removal. (D) All decay removed [5].

In today's rapidly advancing technological landscape, where automation is increasingly integrated into daily life, dental caries continues to be one of the most widespread chronic diseases, affecting people of all ages. A notable gap exists at the interface of oral health and technology, there are no mobile applications specifically designed to detect dental cavities early on. The majorities of caries detection technologies currently in use is either integrated into bigger, frequently costly dental equipment or still rely on conventional methods. In particular areas where access to traditional dental services is limited, the creation of a mobile application that uses cutting-edge technology, such as deep learning algorithms, to identify early signs of dental caries could transform preventive dental care and make it more affordable and accessible.

Table I: Earlier Researches about the Caries Detection Methods

| S. # | Name of paper                                                                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | AI Model Category | Limitation                                                                                                                                                                                                                                                                                                                                                   |
|------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Detection and diagnosis of dental caries using a deep learning-based convolutional neural network algorithm [6] | <p>This study evaluated the effectiveness of deep convolutional neural networks (CNNs) in detecting and diagnosing dental caries using periapical radiographs. The CNN model achieved an accuracy of 89.0% for premolars and 88.0% for molars, with sensitivities of 84.0% and 92.3% and specificities of 94.0% and 84.0%, respectively.</p> <p>The results showed that CNNs are more promising in improving the accuracy and efficiency of dental caries detection in clinical settings.</p>                                                                                                                                      | DL                | The study used a specific deep CNN architecture (GoogLeNet Inception v3) for analysis but did not explore its performance with different datasets or in real-world situations.                                                                                                                                                                               |
| 2    | Automated caries detection with smartphone color photography using machine learning [7]                         | <p>This study used smartphone images to detect tooth decay on chewing surfaces based on the ICDAS system. Researchers photographed 620 unrestored molars and premolars, classifying them into three categories: "No Surface Change" (NSC), "Visually Non-Cavitated" (VNC), and "Cavitated" (C). A Support Vector Machine (SVM) was applied in two steps: distinguishing "C" from "VNC + NSC," and separating "VNC" from "NSC."</p> <p>The best model had 92.37% accuracy, 88.1% sensitivity, and 96.6% specificity for the first classification, and 83.33% accuracy, 82.2% sensitivity, and 66.7% specificity for the second.</p> | ML                | Promising results, but limited by small dataset, imbalanced data, and real-world challenges (e.g. lighting, dental plaque, stains, shadows), potential population-specific effects, and limited interpretability of the SVM model. The study showed that smartphone-based caries detection can be cost-effective despite its limitations like small dataset. |
| 3    | Deep Learning for Caries Detection and Classification [8]                                                       | <p>This study applied deep learning to detect caries lesions and classify their depths (D1, D2, D3) in dental panoramic radiographs. nnU-Net achieved IoU and Dice coefficients of 0.785 and 0.663, respectively, with accuracy and recall rates of 0.986 and 0.821. DenseNet121 achieved accuracies of 0.957 for D1, 0.832 for D2, and 0.863 for D3 lesions.</p> <p>A total of 1160 dental panoramic films were evaluated.</p>                                                                                                                                                                                                    | DL                | It relied on data from a single institution, excluded blurred images, lacked a gold standard for labeling caries lesions, did not include external validation.                                                                                                                                                                                               |
| 4    | Deep Learning Application in Dental Caries Detection Using                                                      | This study applied deep learning algorithms to diagnose caries using smartphone images. The dataset consisted of 1902 photos from 695 people, taken with                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | DL                | The study's models demonstrated low sensitivity in detecting early-stage caries, likely due to a small sample size,                                                                                                                                                                                                                                          |

|   |                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |    |                                                                                                                                                                                                                                                                                                                                                |
|---|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | Intra oral Photos Taken by Smartphone [9]                                                                   | an iPhone 7. Four models Faster R-CNN, YOLOv3, RetinaNet, and SSD were tested. YOLOv3 and Faster R-CNN showed the highest sensitivity for cavitated caries (87.4% and 71.4%, respectively). However, their sensitivity for visually non-cavitated (VNC) lesions was lower (36.9% and 26%, respectively). Specificity was above 86% for cavitated caries and above 71% for VNC.                                                                                                                                                                                  |    | variations in image quality, and the need for further optimization.                                                                                                                                                                                                                                                                            |
| 5 | Caries detection with tooth surface segmentation on intra oral photographic images using deep learning [10] | This study evaluates a deep learning algorithm for detecting caries by segmenting tooth surfaces in intraoral photos. Using 2,348 images from 445 participants, this research applied CNN models (U-Net, ResNet-18, and Faster R-CNN) for classification and localization.<br><br>The models achieved an accuracy of 0.813 and an area under the ROC curve of 0.837. It also showed improved localization performance, with a sensitivity of 0.889 and an average precision of 0.868 for caries detection.                                                      | DL | The Limitations such as image quality variations, dataset size, and the need for clinical validation are important factors to consider while developing any AI-powered system in the medical field.                                                                                                                                            |
| 6 | Dental Caries Detection Using Faster R-CNN and YOLO V3 [11]                                                 | The paper employs deep learning to detect dental cavities in live images captured by a camera. The dataset comprises 300 images, with 80% used for training and 20% for testing. Faster R-CNN achieved an accuracy of 80%, outperforming YOLO V3, which achieved 75%.                                                                                                                                                                                                                                                                                           | DL | This study has limitation, it uses small dataset size of roughly 300 images for training and There were also challenges in capturing posterior teeth due to image angles.                                                                                                                                                                      |
| 7 | Dental Caries Detection and Classification in CBCT Images Using Deep Learning [12]                          | This study evaluates the use of deep learning for diagnosing tooth caries in cone beam computed tomography (CBCT) images. It included a dataset of 382 molar teeth with caries and 403 non-carious molars. A multiple-input convolutional neural network (CNN) was used for detection and classification.<br><br>The results showed a diagnostic accuracy of 95.3% for carious molars and 94.8% for non-carious ones, with sensitivity, specificity, and F1 scores all above 92%. These results demonstrate the model's high performance in dental diagnostics. | DL | The dataset used may lack representation of diverse tooth caries variations, potentially hindering the model's ability to generalize to real-world scenarios.<br><br>Results generated by the algorithms might be challenging for clinicians to interpret, limiting their confidence in utilizing the technology for informed decision-making. |

It can be concluded from Table 1 that no existing mobile application has been implemented for early caries assessment. Our research addresses this gap by proposing a smartphone application designed to meet the urgent need for an accessible and user-friendly tool that supports early caries detection. Our research proposes a smartphone application to meet the urgent demand for an approachable and user-friendly tool that supports the early diagnosis of cavities. Our approach fills the gap left by previous technologies by incorporating the capacity to evaluate the degree of tooth decay and easily schedule appointments with surrounding dental offices. Moreover, we aim to overcome the limitations of prior studies, such as restricted datasets, poor generalization, and challenges in detecting early-stage caries, by introducing deep learning models that were not explored in previous research. Our application not only addresses these gaps but also empowers users to monitor their oral health proactively, supports preventive care, and facilitates timely decision-making through enhanced diagnostic precision and usability.

### III. METHODOLOGY

#### A. PROPOSED SOLUTION

We propose to create the AI powered Dental Caries Detection application in order to address the issues of expense, time limits, and restricted accessibility in traditional dental care. With the use of Deep learning state-of-the-art models, like YOLO (You Only Look Once) [1], EfficientDet [2], Mobilenet [3], and ResNet-50 [4], this innovative approach enables people to recognize and effectively treat dental caries.

#### B. PROPOSED FRAMEWORK

Our proposed framework for implementing the AI powered dental caries detection is shown in Fig. 2 and consists of four main functionalities namely (1) *Caries Assessment* (2) *Locate Dentist* (3) *Personalized Health Plan* (4) *Educational Resources*. These are further defined as follows;

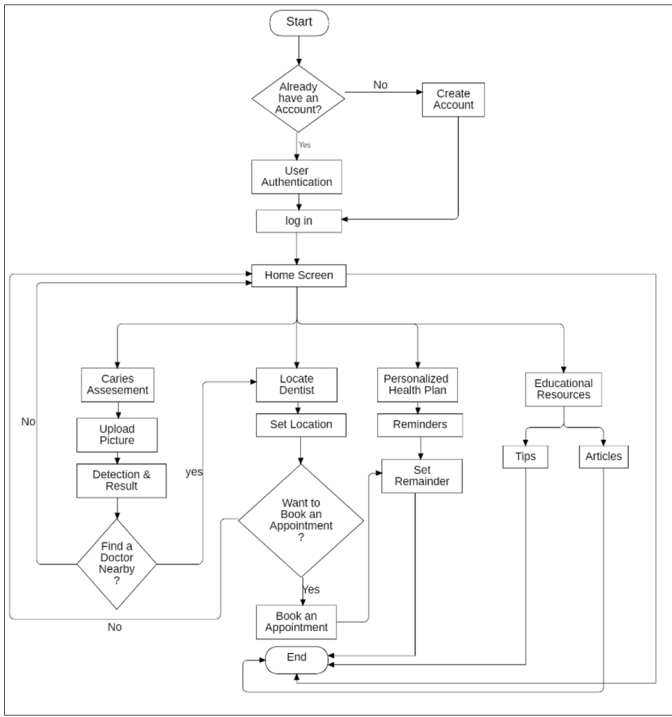


Fig. 2. The Flowchart for Proposed AI powered Dental Caries Detection

**1. Caries Assessment-** Users are first prompted to share pictures of their teeth using the Caries Assessment function, which is simple to use with a smartphone camera. The program guides users through this procedure, making sure the photos are sharp and capture all the angles required for precise analysis. The application uses modern computer vision algorithms to carry out detection and result generation after the photographs are submitted. The software provides a smooth transition to the Find a Dentist function, where users may find and schedule appointments with nearby dentists, if the assessment identifies possible problems. Users may book an appointment for quick and easy care right within the app, which lists available dentists along with their profiles and current availability.

**2. Locate Dentist-** The Locate Dentist feature of the app begins with the Locate Dentist step, where users can search for nearby dental professionals. Users can then set their location either manually or by enabling GPS to ensure accurate results. Once the location is set, the app displays a list of available dentists in the area, complete with profiles and contact information. The next query posed to users is, "Do you want an appointment?" If they answer in the yes, they can use the app to schedule an appointment by selecting a time and date that works for them. To make sure the user doesn't forget the appointment, the app offers to set a reminder after the user bookings. If the user chooses not to, the app's home screen is automatically displayed so they can continue exploring its other features.

**3. Personalized Health Plans-** The app's Personalized Health Plans feature allows users to receive tailored dental care recommendations based on their specific needs and conditions. After an assessment, the app generates a customized health plan that may include preventive care routines, dietary suggestions, and regular check-ups. To help

users stay on track with their dental care, the app also offers the ability to set reminders for key activities outlined in their personalized health plan.

**4. Educational Resource -** The app's Educational Resources feature serves as a valuable repository of information to help users enhance their dental knowledge and maintain good oral hygiene. Within this section, users can access a wide range of tips and articles that cover various aspects of dental health. From daily oral care routines to understanding the causes and prevention of common dental issues, these resources are designed to be both informative and easy to understand.

## IV. IMPLEMENTATION

### A. DATASET

We processed a comprehensive dataset Dental mate caries detection [13] which includes RGB images representing common caries locations and positions. This involved annotating and categorizing the dataset into relevant classes to facilitate model training. The dataset contains many classes such as black stain, decay, caries, tooth-decay and many more and it contains 7447 images, each with a resolution of 640 x 640 pixels. This ensures coverage of caries across different locations. Some of the images from the dataset are shown below in the Figs. 3-6.

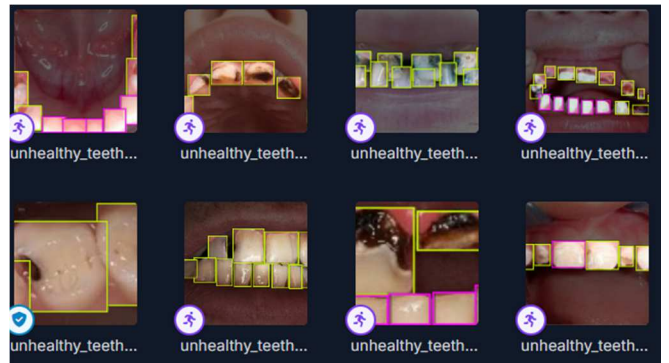


Figure 3 Sample Images of Cavity Class from the Benchmark Dataset [13]



Figure 4 Sample Images of Healthy Tooth class from Benchmark Dataset [13]

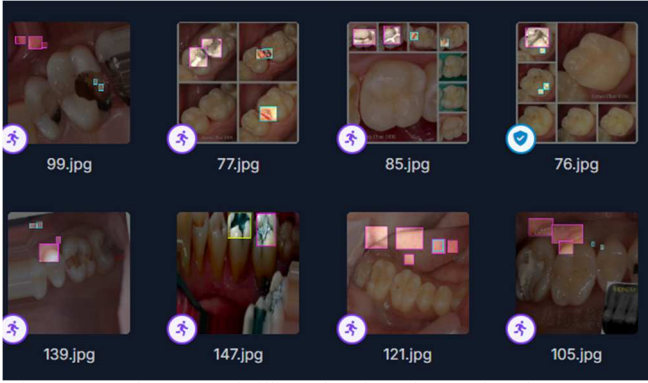


Figure 5 Sample Images of Filling Class from the Benchmark Dataset [13]

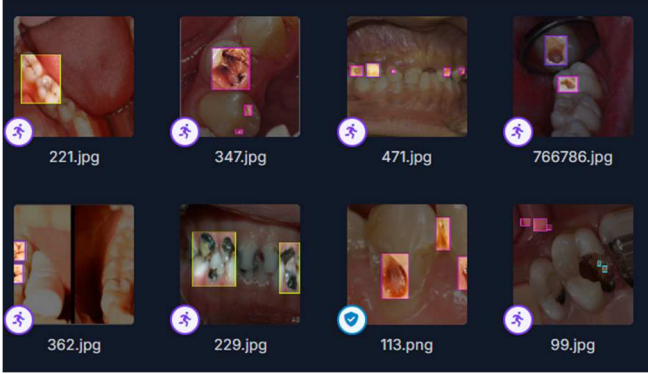


Figure 6 Sample Images of Decay Class from the Benchmark Dataset [13]

## B. MODEL TRAINING

We have trained and tested the model on YOLOv8 [14], ResNet 50 [4], EfficientDet [2], and MobileNet [3]. To ensure effectiveness and repeatability of results, we conducted the training and testing across these four models multiple times, using a split size of 70% for the training set (5,213 images), 19% for the validation set (1,415 images), and 11% for the test set (819 images), and computed the average performance. The results obtained from training and testing on YOLOv8 [14], EfficientDet [2], MobileNet [3], and ResNet 50 [4], along with their average values, are depicted in Table II and explained in the Results Section V ahead. The best values are highlighted in bold.

## C. EVALUATION PARAMETERS

The models are evaluated based on mean average precision (mAP), average recall, and total loss.

# V. RESULTS AND DISCUSSION

Table II Performance Evaluation of the State-of-the-art DL Models for Dental Caries Detection

| Model                | Steps/Epochs        | mAP<br>@[IoU=0.50:0.95] | mAP<br>@[IoU=0.50] | Average<br>Recall | Total<br>Loss |
|----------------------|---------------------|-------------------------|--------------------|-------------------|---------------|
| MobileNet V2 [3]     | 25,000 Steps        | 0.433                   | 0.646              | 0.544             | 0.243         |
| <b>ResNet 50 [4]</b> | <b>25,000 Steps</b> | <b>0.436</b>            | <b>0.670</b>       | <b>0.550</b>      | <b>0.251</b>  |
| EfficientDet D1 [2]  | 25,000 Steps        | 0.323                   | 0.457              | 0.496             | 0.428         |
| YOLOv8 [14]          | 100 Epochs          | 0.416                   | 0.571              | 0.582             | 2.03          |

Based on the comparison of the four models in Table 2, ResNet-50 achieved the highest mAP @[IoU=0.50:0.95] of 0.436 and mAP @[IoU=0.50] of 0.670. It also had the highest recall of 0.550, suggesting that it could correctly identify a higher proportion of true positives compared to the other models. The total loss was relatively low at 0.251, indicating efficient training and learning. MobileNet V2 had a slightly lower mAP@[IoU=0.50:0.95] of 0.433 and mAP@[IoU=0.50] of 0.646 but maintained a strong recall of 0.544. Its total loss was the lowest among the models at 0.243, which reflects better optimization and potentially faster convergence. EfficientDet D1 showed the lowest mAP@[IoU=0.50:0.95] of 0.323 and mAP@[IoU=0.50] of 0.457, indicating that it struggled with overall detection accuracy. However, it had a decent recall of 0.496, showing

that it was still effective in identifying a significant portion of the objects. Its total loss was the highest at 0.428, which may suggest difficulties in training and achieving optimal performance. YOLOv8 performed well with a mAP@[IoU=0.50:0.95] of 0.416 and mAP@[IoU=0.50] of 0.571. It achieved the highest recall of 0.582, which is crucial for applications requiring high sensitivity. However, the total loss was substantially higher at 2.03, which might indicate less efficient training or a higher degree of complexity in the model.

ResNet-50 emerges as the best overall performer considering its balance of high mAP, strong recall, and low total loss. While MobileNet V2 is also competitive, especially with its low total loss, ResNet-50's superior detection capabilities make it the most effective model for the task. YOLOv8 shows

promise with high recall but is hindered by its high total loss, making it less efficient in terms of optimization. EfficientDet D1, although good at detecting smaller objects, falls short overall due to its lower accuracy and higher loss.

## VI. CONCLUSION AND FUTURE WORK

In this research, we focused on leveraging AI to automate the dental caries detection for preventive and early care. We utilized four state-of-the-art DL models (YOLOv8, ResNet, EfficientDet, and MobileNet) on benchmark dental caries dataset to evaluate the feasibility of our proposed approach. The results showed that ResNet model performed the best with the highest mAP and recall values while maintaining the lowest total loss. Our results depict the feasibility of AI powered models for dental caries detection.

Future research will concentrate on building a real-time clinical application, further refining the models or using new Data learning models and enlarging the dataset to cover a broader spectrum of dental problems and demographics can increase accuracy. Furthermore, it will be crucial to carry out longitudinal research to assess the models' effectiveness over time in real-world scenarios. To guarantee that AI-driven diagnostics are applied safely and successfully in clinical settings, it will also be essential to address ethical and regulatory issues. With these actions, the advancement of AI in dentistry healthcare will be facilitated, which could result in better patient results and more widespread industry acceptance of these technologies.

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